

Shan Yang

Research Scientist — Multimodal RL & Reasoning Post-Training

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PROFESSIONAL SUMMARY

- Research scientist focused on **multimodal RL post-training** and visual-language reasoning, with 8+ years across foundation models at Adobe, Amazon, and Google Research
- **Sole-author arXiv 2026** on RL post-training for visual reasoning (**Physics-R1**: GSPO+DAPO, **+18.3 pp** on the held-out olympiad benchmark over Qwen3-VL-8B base); ICML 2026 co-author on test-time reasoning (**+35.8%** Math-500)
- Independent open-source releases at huggingface.co/shanyangmie: PHYSCORP-A (6.4K audited multimodal physics corpus), PHYSOLYM-A (500-problem held-out olympiad eval, 99.8% novel-source); prior first-author NeurIPS / ICCV with **1,500+** combined citations

TECHNICAL SKILLS

RL post-training: GRPO, GSPO, DAPO, DPO, RLHF, verl, MM-Eureka, reward shaping, LLM-as-judge protocols, audit-aware curricula

Multimodal foundation models: vision-language transformers, autoregressive video / diffusion, test-time training, attention-based fusion

Infra: PyTorch, FSDP / DeepSpeed, vLLM, distributed training at 10B+ params, HuggingFace, Croissant 1.0 + RAI dataset packaging

SELECTED PUBLICATIONS

Physics-R1: An Audited Olympiad Corpus and Recipe for Visual Physics Reasoning

S Yang

arXiv 2026 (2605.14040) — project / code / data

Sole-author. Three-stage corpus audit (Jaccard $\rightarrow$ mxbai $\rightarrow$ Haiku-4.5 judge), 500-problem novel-source held-out olympiad benchmark, and GSPO+DAPO recipe trained at ~ 360 H200-GPU-hours over 3 seeds on Qwen3-VL-8B; +18.3 pp on PhysOlym-A, +15.7 pp on PhysReason. Method follow-up in preparation.

Beyond Test-Time Training: Learning to Reason via Hardware-Efficient Optimal Control

P Wang, S Yang, X Wang, T Xiao, X Liu, C Yu, Y Lou, P Li, Z Wang, MC Lin, R Vidal

ICML 2026

Linear recurrent architecture leveraging optimal control theory for LLM test-time reasoning (+35.8% Math-500, +10% AIME25).

Schema Perception for Robust Video Question Answering

X Wang, S Yang

Tech Report 2025

Schema-of-Thought prompting framework for multimodal LLM robustness and OOD generalization (+28% Claude 3.5, +13.3% GPT-4o).

ViLA: Efficient Video-Language Alignment for Video Question Answering

X Wang, J Liang, CK Wang, K Deng, Y Lou, MC Lin, S Yang

ECCV 2024

ICAR: Image-based Complementary Auto Reasoning

X Wang, A Liang, J Liang, M Lin, Y Lou, S Yang

AAAI 2024

Attention Bottlenecks for Multimodal Fusion

A Nagrani, S Yang, A Arnab, A Jansen, C Schmid, C Sun

NeurIPS 2021 — **800+** citations

AI Choreographer: Music Conditioned 3D Dance Generation with AIST++

*R Li**, *S Yang**, *DA Ross*, *A Kanazawa*

ICCV 2021 — **700+ citations**

Physics-Inspired Garment Recovery from a Single-View Image

S Yang, *Z Pan*, *T Amert*, *K Wang*, *L Yu*, *T Berg*, *MC Lin*

ACM ToG 2018 — **100+ citations**

Modeling Context in Referring Expressions

L Yu, *P Poirson*, *S Yang*, *A Berg*, *T Berg*

ECCV 2016 — **1600+ citations**

RESEARCH IN PROGRESS

Reasoning-Method paper (Physics-R1 follow-up): Novel multimodal RL post-training method: judge-robust correctness signals + audit-aware curricula extending the GSPO+DAPO recipe; in preparation.

Multimodal Reasoning RL agenda: Reinforcement-learning post-training of vision-language reasoning models with public benchmarks and open-source releases; Physics-R1 (arXiv 2026) is the first artifact in this line of work.

ENGINEERING & RESEARCH EXPERIENCE

Adobe Foundry

Staff Applied Scientist

San Jose, CA

March 2026 – Present

- Post-training research for Adobe Foundry, Adobe's family of generative AI foundation models powering Firefly and creative tooling at scale
- Developing post-training techniques (SFT, RLHF/DPO, LoRA, distillation, test-time training) to align foundation models with creator intent and production quality bars

Amazon (Prime Video & Search)

Senior Applied Scientist Tech Lead

Palo Alto & Sunnyvale, CA

September 2021 – March 2026

- **Prime Video (2025–2026):** Led GenAI Live Action Studio (**5 Applied Scientists**), building autoregressive diffusion architectures and training infrastructure for **10B+ parameter** video generation models; advanced controllable generation via test-time training, teacher-diffusion-forcing, and RL post-training
- **Search (2021–2025):** Led **Amazon Video Search** from zero to launch with **\$500M+ annual revenue gain** (3 Applied Scientists, 2 MLEs, 15-person SDE team); pioneered multimodal fusion architecture across video, text, image, and structured data
- Drove multimodal LLM robustness research: Schema-of-Thought framework, **+28% hallucination reduction**; published at ECCV, AAI; filed multiple patents

Google Research

Senior Research Software Engineer

Mountain View, CA

June 2017 – September 2021

- Led foundational research in multimodal generative models with **3 Research Engineers**; drove agenda on music-conditioned 3D motion synthesis and video-audio-motion modeling; first/co-first author publications at ICCV, NeurIPS, ECCV
- Owned 2D/3D pose reconstruction models deployed on GCP at scale; led end-to-end research-to-production

EDUCATION

University of North Carolina at Chapel Hill

Ph.D., Computer Science

Chapel Hill, NC

2012 – 2017

Shanghai Jiao Tong University

B.E., Software Engineering

Shanghai, China

2008 – 2012